

Generation Interconnection System Impact Study Report

For Queue Project Q22-2116

Beloit Gateway 100 kV

August 15, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2116. The Interconnection Customer proposes a 57.5 MW solar photovoltaic facility interconnecting at the Beloit Gateway 100 kV substation in the Ohio area. The request was received on June 16, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 669 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 23 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$26,501,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
119Th Street – Briarcliff 100 KV	Imperial – Eufaula 500 KV L	200.5	200	10.0
Beach Grove – Dollarhide Substation 100 KV	Beaver – Eufaula 500 KV	191.4	133	39.2
John P Madgett – Beloit Gateway 100 KV	Beavr – Eufaula 500 KV	185.9	171	56.8
Gallatin Tn – Don Julio-Lorac Vista 100 KV	Beaver – Eufaula 500 KV	168.1	120	10.5
John P Madgett – Dollarhide 100 KV	Beaver – Eufala 500 KV Line	167.1	247	50.4
Fort GarInd – Briarcliff 100 KV	Imperial – Eufaula 500 KV	162.4	267	10.0
Beach Grove – Fort Garland 100 KV Line	Imperial – Eufaula 500 KV L	152.2	225	15.4
Velasco – Ix Center 100 KV	Eufaula – Imperial Substation 500 KV	149.3	150	5.3
Getty – Ix Center 100 KV	Imperial – Eufaula 500 KV	147.7	260	6.8
Velasco – Cedar Hill 100 KV	Eufaula – Imperial 500 KV line	143.0	147	5.3
Loring Power Plant – West Union 100 KV	Eufaula – Imperial 500 KV	138.0	232	6.3
Poust – Fox 100 KV	Eufaula – Iperial 500 KV	136.3	241	6.4
Fowles – Cosumnes 100 KV	Getty Substation – Ix Ctr 100 KV	127.6	180	6.4
West Uion – Wellington 1 100 KV	Eufaula – Imperial 500 KV	125.5	251	5.9
Briarwood – Sierraville 100 KV Line	John P Madgett – Beloit Gateway 100 KV	122.5	132	19.1
Fowles – Wellington1 Substation 100 KV	Imperial – Eufaula 500 KV L	119.9	294	5.9
Getty – Cosumnes 100 KV Line	Getty – Ix Center Substation 100 KV	117.8	221	6.4
Don Julio-Lorac Vista – Venture 100 KV	Beaver Substation – Eufaula 500 KV	117.4	160	20.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Cedar Hill – Fox 100 KV	Imperial – Eufaula 500 KV	116.9	303	5.3
Winger – Thayer North 230 KV Line	Imperial Substation – Eufaula 500 KV	105.9	471	6.3
Briarwood – Avon Lake 100 KV	John P Madgett – Beloit Gateway 100 KV L	104.3	106	19.1
Parish – Dollarhide Substation 100 KV	Beaver – Eufaula 500 KV L	102.4	132	11.2
Winger – Juncos Tc 230 KV	Imperial – Eufaula 500 KV	100.5	545	6.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
119Th Street – Briarcliff 100 KV	Rebuild line with higher-capacity conductor	\$1,055,000	100% (\$1,055,000)
Beach Grove – Dollarhide Substation 100 KV	Rebuild line with higher-capacity conductor	\$2,860,000	100% (\$2,860,000)
John P Madgett – Beloit Gateway 100 KV	Rebuild line with higher-capacity conductor	\$3,274,000	100% (\$3,274,000)
Gallatin Tn – Don Julio-Lorac Vista 100 KV	Rebuild line with higher-capacity conductor	\$3,569,000	33% (\$1,188,000)
John P Madgett – Dollarhide 100 KV	Rebuild line with higher-capacity conductor	\$1,772,000	75% (\$1,334,000)
Fort Garland – Briarcliff 100 KV	Rebuild line with higher-capacity conductor	\$2,892,000	100% (\$2,892,000)
Ft Garland – Beach Grove 100 KV Line	Rebuild line with higher-capacity conductor	\$3,439,000	34% (\$1,187,000)
Velasco – Ix Center 100 KV	Rebuild line with higher-capacity conductor	\$2,167,000	4% (\$86,000)
Getty – Ix Center 100 KV	Rebuild line with higher-capacity conductor	\$3,328,000	8% (\$282,000)
Velasco – Cedar Hill 100 KV	Rebuild line with higher-capacity conductor	\$2,322,000	4% (\$93,000)
Loring Power Plant – West Union 100 KV	Rebuild line with higher-capacity conductor	\$2,284,000	26% (\$601,000)
Poust – Fox 100 KV	Rebuild line with higher-capacity conductor	\$3,887,000	6% (\$241,000)
Fowles – Cosumnes 100 KV	Rebuild line with higher-capacity conductor	\$3,588,000	9% (\$326,000)
West Union – Wellington 1 100 KV	Rebuild line with higher-capacity conductor	\$2,348,000	16% (\$374,000)
Briarwood – Sierraville 100 KV Line	Rebuild line with higher-capacity conductor	\$2,392,000	100% (\$2,392,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Wellington 1 – Fowles 100 KV	Rebuild line with higher-capacity conductor	\$2,023,000	16% (\$323,000)
Getty – Cosumnes 100 KV	Rebuild line with higher-capacity conductor	\$1,275,000	9% (\$116,000)
Venture – Don Julio-Lorac Vista 100 KV line	Rebuild line with higher-capacity conductor	\$2,727,000	100% (\$2,727,000)
Cedar Hill – Fox 100 KV	Rebuild line with higher-capacity conductor	\$1,720,000	3% (\$50,000)
Winger – Thayer North 230 KV Line	Reconductor line	\$2,451,000	100% (\$2,451,000)
Briarwood – Avon Lake 100 KV	Reconductor line	\$1,360,000	100% (\$1,360,000)
Parish – Dollarhide Substation 100 KV	Reconductor line	\$1,088,000	100% (\$1,088,000)
Winger – Juncos Tc 230 KV	Reconductor line	\$1,862,000	11% (\$201,000)
	Total allocated to project		\$26,501,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.